

Prospective Multicenter Study Evaluating a Combined Circulating Tumor DNA and Circulating Tumor RNA Liquid Biopsy in Metastatic Non–Small Cell Lung Cancer (LIQUIK)

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ABSTRACT

PURPOSE To evaluate the performance of a circulating tumor DNA (ctDNA) and circulating tumor RNA (ctRNA) liquid biopsy, LiquidHALLMARK (LHM), compared with tissue next-generation sequencing (NGS) and Guardant360 CDx (G360 ctDNA) liquid biopsy for biomarker detection in metastatic nonsquamous non–small cell lung cancer.

PATIENTS AND METHODS This multicenter, prospective study (ClinicalTrials.gov identifier: [NCT04703153](#)) enrolled patients across the United States and Singapore. Patients were tested with tissue NGS, LHM, and G360 ctDNA. The primary objective was noninferiority of LHM ctDNA to tissue NGS and G360 ctDNA. Secondary analyses included turnaround time (TAT), overall response rate (ORR), and progression-free survival (PFS), with exploratory analysis of the clinical utility of ctRNA.

RESULTS LHM ctDNA (48.2%) detected 11.4% fewer biomarker-positive patients than tissue NGS (59.6%) and did not meet noninferiority criteria. Compared with tissue NGS, LHM ctDNA and G360 ctDNA were concordant in 72.1% and 66.1% of patients, establishing noninferiority of LHM ctDNA to G360 ctDNA ($P = .002$). TAT was shorter for LHM ctDNA than for tissue NGS (mean 9.7 v 21.7 days; $P < .001$). ORR/PFS was similar in patients receiving targeted therapy based on all three assays. Addition of ctRNA increased the diagnostic yield of tissue NGS–confirmed gene rearrangements by 28.6% relative to LHM ctDNA and all actionable biomarkers by 15.6% relative to G360 ctDNA. LHM ctDNA and ctRNA (51/68) detected 8.8% more biomarker-positive patients than G360 ctDNA (45/68), demonstrating superiority of LHM ctDNA and ctRNA ($P = .001$).

CONCLUSION LHM ctDNA is noninferior to G360 ctDNA, but not tissue NGS. Treatment outcomes based on liquid biopsy are comparable with those based on tissue NGS. Incorporation of ctRNA into LHM ctDNA improves the diagnostic yield of actionable, tissue NGS–confirmed gene rearrangements.

ACCOMPANYING CONTENT

-  [Data Sharing Statement](#)
-  [Data Supplement](#)
-  [Protocol](#)

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INTRODUCTION

Lung cancer is the most common cancer and the leading cause of cancer mortality worldwide,¹ with non–small cell lung cancer (NSCLC) accounting for nearly 85% of all lung cancers.² Although the 5-year survival rate of NSCLC remains poor at 23%,² substantial progress in the treatment of advanced nonsquamous NSCLC has been made owing to the development of driver mutation–specific targeted therapies over the last decade. Response rates to these targeted therapies range from 50% to 80%, and survival rates in

patients treated with the appropriate US Food and Drug Administration (FDA)–approved targeted therapy have significantly increased.³

Today, several clinical guidelines endorse genomic profiling of advanced nonsquamous NSCLC to identify targetable molecular alterations for appropriate treatments.^{4,5} The latest National Comprehensive Cancer Network (NCCN) NSCLC v3.2025 guidelines endorse the analysis of 11 genes (*EGFR*, *ALK*, *RET*, *ROS1*, *NTRK1/2/3*, *ERBB2*, *KRAS*, *BRAF*, *MET*, *NRG1*, *FGFR2/3*) via a broad panel-based approach using

CONTEXT

Key Objective

To prospectively evaluate how an amplicon-based next-generation sequencing (NGS) liquid biopsy assay that assesses both circulating tumor DNA (ctDNA) and circulating tumor RNA (ctRNA) performs in comparison with tissue-based NGS and ctDNA-only liquid biopsy in identifying guideline-recommended alterations in advanced nonsquamous non–small cell lung cancer.

Knowledge Generated

To our knowledge, this is the first prospective study to evaluate the performance of combined ctDNA and ctRNA liquid biopsy. It demonstrates that combined ctDNA and ctRNA testing is superior to ctDNA testing alone and that although such an approach is still inferior to tissue NGS, liquid biopsy is faster and more accessible than tissue NGS.

Relevance

This study supports the incorporation of ctRNA testing into liquid biopsy, particularly for the detection of actionable gene rearrangements. It further supports the use of combined ctDNA and ctRNA liquid biopsy to complement tissue testing in clinical practice.

next-generation sequencing (NGS).⁶ It further supports the use of RNA-based assays for the detection of gene rearrangements such as *ALK*, *RET*, *ROS1*, *NTRK1/2/3* gene fusions and *MET* exon 14 skipping events.⁶

Although tissue biopsy remains the gold standard for molecular analysis, guidelines increasingly support the use of plasma cell-free DNA/circulating tumor DNA (ctDNA) to complement tissue testing, particularly when insufficient tissue is obtained.^{5,7} An estimated 15% to 40% of NSCLC cases do not undergo complete molecular testing due to insufficient tissue.⁸

Conversely, the use of plasma ctDNA, shed by tumor cells into blood plasma, in liquid biopsies presents a minimally invasive, faster alternative to traditional tissue biopsy.^{7,9} Two NGS-based assays, Guardant360 CDx and FoundationOne Liquid CDx, are currently approved by the FDA as companion diagnostics for cancer molecular profiling.¹⁰ Despite these potential advantages, several limitations hamper the widespread implementation of liquid biopsies in clinical practice, most notably concerns around the lower sensitivity of liquid biopsies and the lack of standards on the recommended analytical performance of liquid biopsies.^{5,7,11} Hence, it remains imperative to evaluate the clinical performance of individual liquid biopsies and to investigate additional strategies that may reduce liquid biopsy false-negative rates, such as incorporating circulating tumor RNA (ctRNA) to improve detection of gene rearrangements.

In this study, the performance of an amplicon-based liquid biopsy that integrates both ctDNA and ctRNA for biomarker detection in metastatic nonsquamous NSCLC (LiquidHALLMARK)^{12,13} was compared against tissue NGS and the FDA-approved hybrid capture-based ctDNA-only

Guardant360 CDx liquid biopsy (G360 ctDNA).¹⁴ The primary aim of the study was to demonstrate the noninferiority of LiquidHALLMARK's ctDNA assay (LHM ctDNA) to tissue NGS and G360 ctDNA. Secondary objectives of the study included analysis of treatment response outcomes in patients with NSCLC based on biomarker detection by tissue NGS, LHM ctDNA, and G360 ctDNA. In addition, exploratory analyses were performed to evaluate the clinical utility of LiquidHALLMARK's ctRNA (LHM ctRNA) assay in detecting guideline-recommended gene rearrangements.

PATIENTS AND METHODS

Study Design and Participants

LIQUIK (LIQUId Biopsy for Detection of Actionable Genomic Biomarkers in Patients With Advanced NSCLC; ClinicalTrials.gov identifier: [NCT04703153](https://clinicaltrials.gov/ct2/show/study/NCT04703153)) was a multicenter, prospective, observational cohort study to compare the performance of the liquid biopsy LHM ctDNA with tissue NGS and G360 ctDNA for detecting guideline-recommended biomarkers in metastatic nonsquamous NSCLC.

The study was approved by the relevant institutional review boards and was conducted in accordance with the provisions of the Declaration of Helsinki, Good Clinical Practice guidelines, and applicable regulatory requirements. All study participants provided written informed consent. This study is reported in compliance with STARD guidelines¹⁵ (Data Supplement, Text S1).

LIQUIK was conducted in 10 centers across the United States and Singapore. Eligible patients were age ≥ 21 years with newly diagnosed, treatment-naïve, histologically or cytologically confirmed, metastatic nonsquamous NSCLC.

Genotyping Procedures

Tissue and liquid biopsy were performed by NGS only and included nine NCCN guideline-recommended biomarkers for NSCLC (G9): *EGFR*, *ALK*, *RET*, *ROS1*, *BRAF*, *KRAS*, *MET*, *ERBB2*, and *NTRK1/2/3*. Two liquid biopsies were used, the FDA-approved G360 ctDNA (Guardant Health, Redwood City, CA) and the Clinical Laboratory Improvement Amendments–certified, College of American Pathologists–accredited NGS LHM ctDNA (Lucence Health, Palo Alto, CA; Lucence Diagnostics, Singapore). Over the study period, LHM underwent a version upgrade to include cell-free RNA (cfRNA) analysis (LHM ctDNA and ctRNA). Patients who reconsented to cfRNA testing and had sufficient plasma also underwent cfRNA testing.

Statistical Analysis

Logistic generalized estimating equation models with bootstrapping were used to estimate differences in detection rate and left-sided 95% CIs.

Secondary analyses of the study included turnaround time (TAT), overall response rate (ORR), and progression-free survival (PFS). Statistical difference in TAT was determined using a linear mixed model. Exact 95% CI was calculated for ORR and log-rank testing was used to determine differences in PFS.

Additional details are provided in the Data Supplement (Text S2).

Human Investigations

The authors declare that the investigators performed the human investigations after approval by respective local Human Investigations Committee or Ethics Committees. In addition, all data are anonymized to protect the identities of individuals involved in the research. The investigators obtained informed consent from each participant or each participant's guardian before their participation in the study. This multicenter, prospective study is registered with the following trial number [NCT04703153](#).

RESULTS

Patients

A total of 151 biopsy-confirmed patients with metastatic nonsquamous NSCLC was enrolled across the United States and Singapore between April 2021 and December 2022 (Table 1). NSCLC subtype was predominantly adenocarcinoma (96.0%). The median age at diagnosis was 68 (range, 31–93). Males and females were equally represented in the cohort (males, 55.0%; females, 45.0%).

Of the 151 enrolled patients, one was unable to provide blood at baseline. In the remaining 150, 36 patients were deemed

TABLE 1. Baseline Demographic and Clinical Characteristics of Study Participants

Characteristic	No. (%)
Total	151 (100)
Sex	
Female	68 (45.0)
Male	83 (55.0)
Age at diagnosis, years, median (range)	68 (31–93)
Race	
Asian	115 (76.2)
White	18 (11.9)
Black or African American	8 (5.3)
Native Hawaiian or other Pacific Islander	4 (2.6)
Unknown	6 (9.3)
Ethnicity	
Hispanic	5 (3.3)
Non-Hispanic	143 (94.7)
Unknown	3 (2.0)
Non–small cell lung cancer subtype	
Adenocarcinoma	145 (96.0)
Mixed tumor	1 (0.7)
Large cell	1 (0.7)
Unclassified	4 (2.6)

ineligible for primary analysis—23 had insufficient tissue for analysis (tissue quantity not sufficient; QNS), tissue NGS could not be performed in another eight patients, G360 ctDNA could not be performed in three patients, and LHM ctDNA could not be performed in two patients. A total of 114 patients with all three biomarker assays performed were included in the primary analysis (Fig 1).

Molecular Profiling Outcomes

Of the 114 patients, tissue NGS detected a G9 biomarker in 68 patients (59.6%), whereas LHM ctDNA detected a G9 biomarker in 55 patients (48.2%). Forty-nine patients were G9-positive by both methods, whereas 19 were positive by tissue NGS only and six by LHM ctDNA only, giving an overall sensitivity of 72.1%, specificity of 87.0%, and accuracy of 78.1% (Data Supplement, Table S1). Across each G9 category, the overall accuracy ranged from 93.0% to 100%, demonstrating high clinical concordance of LHM ctDNA to tissue NGS (Table 2). On the basis of the –11.4% difference in proportion of G9-positive patients, noninferiority of LHM ctDNA to tissue NGS was not met (left-sided 95% CI, –16.4%; $P = 1.00$).

Of the 68 tissue NGS-confirmed G9-positive patients, G360 ctDNA detected the same G9 biomarker in 45 patients (66.1%), whereas LHM ctDNA detected the same G9 biomarker in 49 patients (72.1%), giving a 5.9% difference and demonstrating noninferiority of LHM ctDNA to G360 ctDNA (left-sided 95% CI, 0%; $P = .002$). Positive percent

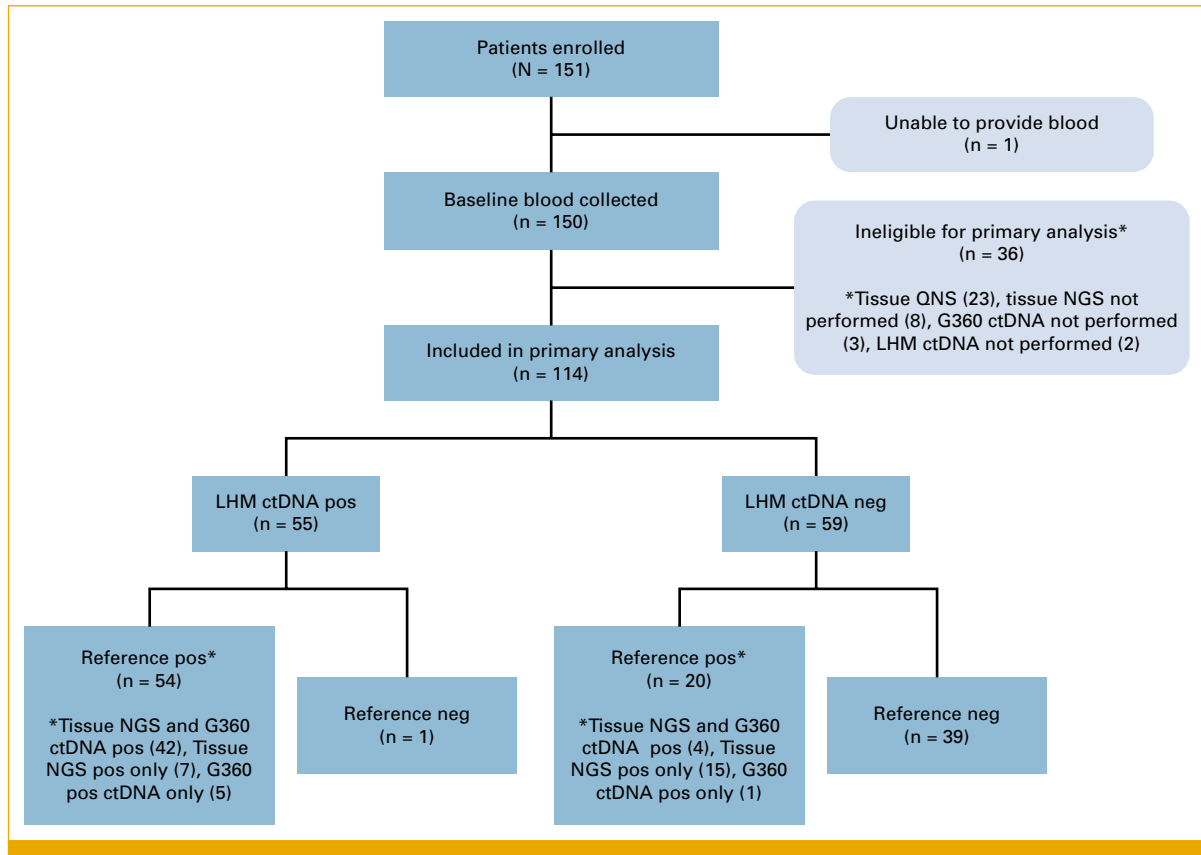


FIG 1. STARD diagram of the LIQUIK study population and the biomarker detection rates of each assay. ctDNA, circulating tumor DNA; G360 ctDNA, Guardant360 CDx; LHM ctDNA, LiquidHALLMARK ctDNA; neg, negative; NGS, next-generation sequencing; pos, positive; QNS, quantity not sufficient.

agreement between LHM ctDNA and G360 ctDNA was 93.3%, negative percent agreement was 69.6%, and overall percent agreement (OPA) was 85.3%, with LHM ctDNA and G360 ctDNA detecting seven and three G9-positive patients missed by the other assay, respectively. OPA across all G9 categories was high (range, 92.6%-100%; Fig 2).

In the 46 patients deemed G9-negative by tissue NGS, seven patients had a G9 biomarker (two *ALK*, two *EGFR*, two *KRAS*, one *ERBB2*) detected by either liquid biopsy. Five (71.4%) of the seven patients had the same biomarker detected by both liquid biopsies, whereas LHM ctDNA and G360 ctDNA each uniquely detected one G9-positive patient (Data Supplement, Table S2).

In total, tissue NGS could not be performed in 20.5% (31/151) of patients, including 23 patients with tissue QNS. Among these, LHM ctDNA and G360 ctDNA detected 15 G9-positive patients, with 11 (73.3%) of 15 detected by both assays. LHM ctDNA and G360 ctDNA each uniquely detected two G9-positive patients (Data Supplement, Table S3).

When considering the entire study cohort, tissue NGS detected a G9 biomarker in 70 (46.4%) of 151 patients, whereas

LHM ctDNA detected a G9 biomarker in 69 (45.7%) of 151 patients, which also did not meet the criteria for non-inferiority (difference, -0.7%; left-sided 95% CI, -5.6%; $P = .38$). G360 ctDNA detected a G9 biomarker in 66 (43.7%) of 151 patients. The biomarker prevalence from each assay is shown in Figure 3A and the overlap in all G9 biomarkers and gene rearrangements identified by each assay is shown in Figure 3B and the Data Supplement (Fig S1). A full list of patient-level findings can be found in the Data Supplement.

The mean TAT of LHM ctDNA and G360 ctDNA from the time of blood draw to receipt of the report was 9.7 days (range, 5-37 days) and 10.1 days (range, 6-18 days), respectively. The mean TAT of tissue NGS from the time of tissue collection to receipt of the report was 21.7 days (range, 3-98 days). The mean TAT was significantly different between LHM ctDNA and tissue NGS ($P < .001$), but not between LHM ctDNA and G360 ctDNA ($P = .63$).

Treatment Outcomes

Of the 151 enrolled patients, 129 were put on first-line therapy following biomarker testing, with 64 (49.6%) on targeted therapy, 53 (41.1%) on chemotherapy, and 39

TABLE 2. Concordance of LHM ctDNA With Tissue NGS for the Detection of Guideline-Recommended Biomarker-Positive Patients in Metastatic Nonsquamous Non–Small Cell Lung Cancer

Actionable Biomarker	LHM ctDNA Findings	Tissue NGS Pos	Tissue NGS Neg	Total	Sens	Spec	PPV	NPV	Acc
<i>EGFR</i> L858R or exon 19 deletion	LHM pos	28	2	30	82.4%	97.5%	93.3%	92.9%	93.0%
	LHM neg	6	78	84					
	Total	34	80	114					
<i>EGFR</i> exon 20 mutation	LHM pos	5	0	5	83.3%	100%	100%	99.1%	99.1%
	LHM neg	1	108	109					
	Total	6	108	114					
<i>KRAS</i> G12C	LHM pos	4	1	5	100%	99.1%	80.0%	100%	99.1%
	LHM neg	0	109	109					
	Total	4	110	114					
<i>ALK</i> rearrangement	LHM pos	3	2	5	50.0%	98.1%	60.0%	97.2%	95.6%
	LHM neg	3	106	109					
	Total	6	108	114					
<i>ERBB2</i> (HER2) mutation	LHM pos	3	1	4	60.0%	99.1%	75.0%	98.2%	97.4%
	LHM neg	2	108	110					
	Total	5	109	114					
High-level <i>MET</i> amplification	LHM pos	2	0	2	66.7%	100%	100%	99.1%	99.1%
	LHM neg	1	111	112					
	Total	3	111	114					
<i>MET</i> exon 14 skipping	LHM pos	2	1	3	40.0%	99.1%	66.7%	97.3%	96.5%
	LHM neg	3	108	111					
	Total	5	109	114					
<i>RET</i> rearrangement	LHM pos	1	0	1	100%	100%	100%	100%	100%
	LHM neg	0	113	113					
	Total	1	113	114					
<i>BRAF</i> V600E	LHM pos	1	1	2	50.0%	99.1%	50.0%	99.1%	98.2%
	LHM neg	1	111	112					
	Total	2	112	114					
<i>ROS1</i> rearrangement	LHM pos	1	0	1	25.0%	100%	100%	97.3%	97.4%
	LHM neg	3	110	113					
	Total	4	110	114					

Abbreviations: Acc, accuracy; ctDNA, circulating tumor DNA; LHM ctDNA, LiquidHALLMARK ctDNA; NGS, next-generation sequencing; NPV, negative predictive value; PPV, positive predictive value; Sens, sensitivity; Spec, specificity

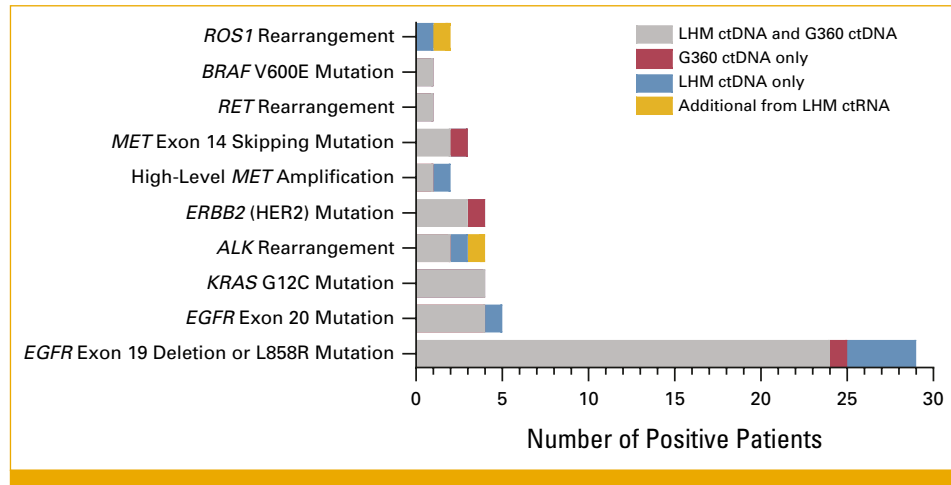


FIG 2. Concordance of LHM with G360 for the detection of tissue NGS–confirmed guideline-recommended biomarker-positive patients in metastatic nonsquamous NSCLC. ctDNA, circulating tumor DNA; ctRNA, circulating tumor RNA; G360 ctDNA, Guardant360 CDx; LHM ctDNA, LiquidHALLMARK ctDNA; LHM ctRNA, LiquidHALLMARK ctRNA; NGS, next-generation sequencing; NSCLC, non–small cell lung cancer.

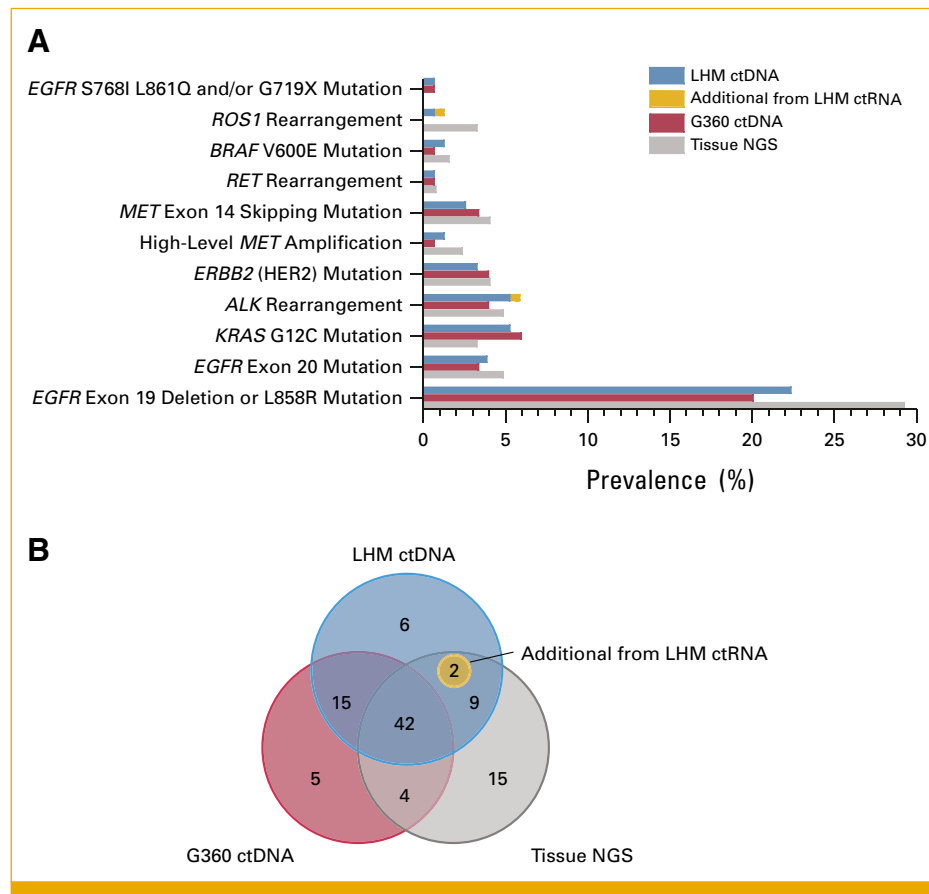


FIG 3. (A) Biomarker prevalence and (B) overlap of biomarkers detected for each assay in the LIQUIK study. ctDNA, circulating tumor DNA; ctRNA, circulating tumor RNA; G360 ctDNA, Guardant360 CDx; LHM ctDNA, LiquidHALLMARK ctDNA; LHM ctRNA, LiquidHALLMARK ctRNA; NGS, next-generation sequencing.

(30.2%) on immunotherapy (including 27 patients on combination therapy). Of the 64 on first-line targeted therapy, 47 had matched biomarker findings based on tissue NGS, 46 based on LHM ctDNA, and 43 based on G360 ctDNA.

A total of 112 patients had tumor assessments at baseline and within 6 months of treatment initiation. On the basis of RECIST v1.1, the ORR of patients on first-line targeted therapy, chemotherapy, and immunotherapy was 40.4% (95% CI, 27.6% to 54.2%), 17.8% (95% CI, 8.0% to 32.1%), and 9.4% (95% CI, 2.0% to 25.0%), respectively. Among patients treated with first-line targeted therapy, ORR was similar between patients where treatment was matched to the biomarker findings based on tissue NGS (45.2% [95% CI, 29.9% to 61.3%]), LHM ctDNA (39.0% [95% CI, 24.2% to 55.5%]), and G360 ctDNA (36.8% [95% CI, 21.8% to 54.0%]).

The median PFS of patients on first-line targeted therapy and chemo/immunotherapy was 23.6 months (95% CI, 15.9 to not reached [NR]) and 3.8 months (95% CI, 2.8 to 5.5), respectively; the PFS of patients on first-line targeted therapy was significantly longer than those not on targeted therapy (hazard ratio, 0.26 [95% CI, 0.16 to 0.41]; $P < .001$; Fig 4A). The median PFS was similar between patients where targeted therapy was matched to the biomarker findings based on tissue NGS (23.6 months [95% CI, 16.3 to NR]), LHM ctDNA (18.6 months [95% CI, 12.0 to NR]), and G360 ctDNA (20.1 months [95% CI, 14.4 to NR]; Fig 4B).

Utility of Plasma ctRNA Testing

In total, 101 patients were analyzed by LHM ctRNA. LHM ctRNA detected seven G9 biomarkers, including six confirmed by tissue NGS and one confirmed by both LHM ctDNA and G360 ctDNA (Table 3). All seven patients were treated

with first-line targeted therapy; six of seven had evaluable RECIST v1.1 response, whereas one case was lost to follow-up. The ORR of LHM ctRNA-positive patients was 66.7% (4/6). Of note, of the two patients with tissue NGS-confirmed gene arrangements detected by ctRNA only, one patient was lost to follow-up, whereas the second, *ROS1*-positive patient placed on entrectinib exhibited a partial response within the first 6 months of treatment (Table 3).

When considering only tissue NGS-confirmed gene rearrangements ($n = 16$), LHM ctRNA detected six (46.2%; three patients had insufficient plasma for ctRNA testing) of 13 biomarkers, whereas LHM ctDNA and G360 ctDNA detected seven (43.8%) of 16 and six (37.5%) of 16 biomarkers, respectively. Collectively, LHM ctDNA and ctRNA detected nine (56.3%) of 16 tissue NGS-confirmed gene rearrangements, with LHM ctRNA increasing the diagnostic yield of gene rearrangements by 28.6% (2/7) relative to LHM ctDNA and 50.0% (3/6) relative to G360 ctDNA. LHM ctDNA and ctRNA detected four and G360 ctDNA detected one gene rearrangement missed by the other assay. OPA between LHM ctDNA and LHM ctRNA for tissue NGS-confirmed gene rearrangements was 64.3% (Data Supplement, Table S4). When not restricted to gene rearrangements, LHM ctDNA and G360 ctDNA detected 50 (71.4%) of 70 and 45 (64.3%) of 70 tissue NGS-confirmed G9 biomarkers, respectively, whereas LHM ctDNA and ctRNA detected 52 (74.3%) of 70 G9 biomarkers. Hence, LHM ctRNA increased the diagnostic yield of G9 biomarkers by 4.0% (2/50) relative to LHM ctDNA and 15.6% (7/45) relative to G360 ctDNA.

Overall, inclusion of ctRNA increased the G9 biomarker sensitivity of LHM relative to tissue NGS from 72.1% (49/68) to 75.0% (51/68) (Fig 2 and Data Supplement, Table S5), compared with 66.2% (45/68) in G360 ctDNA (Data

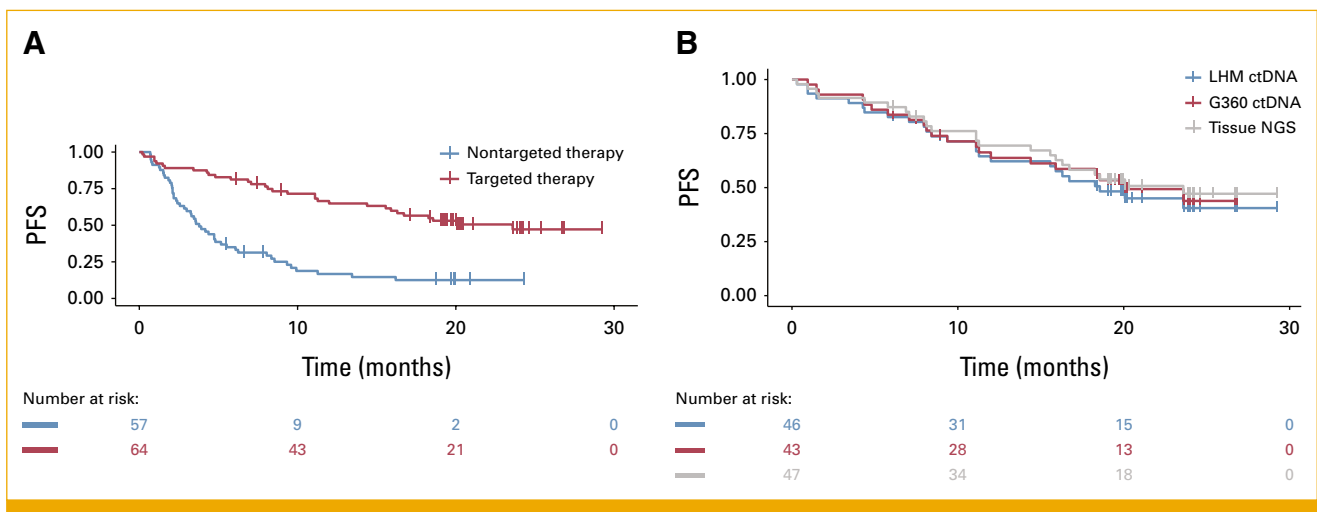


FIG 4. Progression-free survival of (A) patients treated with targeted therapy versus nontargeted therapy in the evaluable cohort and (B) patients treated with targeted therapy based on matched biomarker finding in LHM ctDNA, G360 ctDNA, and Tissue NGS. ctDNA, circulating tumor DNA; G360 ctDNA, Guardant360 CDx; LHM ctDNA, LiquidHALLMARK ctDNA; NGS, next-generation sequencing.

TABLE 3. Patients With Gene Rearrangements Detected by LHM ctRNA and Their Response to Treatment

Patient ID	ctRNA Findings	LHM ctDNA	G360 ctDNA	Tissue NGS	Treatment	Best Response
4	<i>MET</i> exon 14 skipping	1	1	1	Capmatinib	PR
8	<i>ALK</i> rearrangement	1	1	0	Alectinib	PR
14	<i>ALK</i> rearrangement	0	0	1	Alectinib	LTFU
21	<i>ALK</i> rearrangement	1	1	1	Alectinib	SD
22	<i>ROS1</i> rearrangement	0	0	1	Entrectinib	PR
39	<i>ROS1</i> rearrangement	1	0	1	Crizotinib	PR
75	<i>ALK</i> rearrangement	1	1	1	Lorlatinib	SD

NOTE. 0 indicates the biomarker was not detected by the assay; 1 indicates the biomarker was detected by the assay.

Abbreviations: ctDNA, circulating tumor DNA; ctRNA, circulating tumor RNA; G360 ctDNA, Guardant360 CDx; LHM ctDNA, LiquidHALLMARK ctDNA; LHM ctRNA, LiquidHALLMARK ctRNA; LTFU, lost to follow-up; NGS, next-generation sequencing; PR, partial response; SD, stable disease.

Supplement, Table S6). LHM ctDNA and ctRNA detected 8.8% more G9-positive patients than G360 ctDNA (left-sided 95% CI, 2.4%; $P = .001$), demonstrating superiority of LHM ctDNA and ctRNA.

DISCUSSION

In this multicenter prospective study, the clinical performance of the amplicon-based liquid biopsy LHM was compared with tissue NGS and the FDA-approved G360 ctDNA liquid biopsy in a cohort of patients with metastatic nonsquamous NSCLC. Although noninferiority of LHM ctDNA to tissue NGS was not met, LHM ctDNA was non-inferior to G360 ctDNA, with LHM ctDNA detecting 5.9% more tissue NGS-confirmed G9-positive patients with NSCLC. Although the results here are in conflict with several previous studies demonstrating noninferiority of liquid biopsy to tissue biopsy,^{16,17} the key difference is the use of tissue NGS in this study as a comparator rather than the physician's choice of standard-of-care tissue test, enabling a more direct comparison between liquid NGS and tissue NGS.

Tissue testing has been considered the gold standard for cancer diagnostics, enabling not just the histopathological analysis of tumors but today also informing therapeutic decisions based on biomarker identification.¹⁸ In this study, successful tissue NGS detected an actionable G9 mutation in nearly 60% of patients. This was, however, reduced to 46.4% when considering the entire cohort, due to 15.2% of patients being ineligible for tissue NGS because of insufficient nucleic acid quantity, highlighting the importance of real-world evaluation. By contrast, >95% of patients were eligible for liquid biopsy analysis. Even with the lesser sensitivity of liquid biopsies (76.5% combined sensitivity of both liquid biopsies relative to tissue NGS), several advantages of liquid biopsies were highlighted. First, liquid biopsy findings were highly specific, with LHM ctDNA demonstrating >98% specificity per G9 biomarker. Second, liquid biopsy identified a G9 biomarker in 45.5% of cases where tissue NGS could not be performed, with a 73.3% concordance between LHM ctDNA and G360 ctDNA. Third,

where tissue NGS did not identify a G9 biomarker, liquid biopsy identified a G9 biomarker in 15.2% of cases, with a 71.4% concordance between LHM ctDNA and G360 ctDNA. Hence, when considering the entire study cohort, tissue biopsy and liquid biopsy detected a comparable number of G9-positive patients (tissue NGS, 46.4%; LHM ctDNA, 45.7%; G360 ctDNA, 43.7%). Importantly, treatment response rates based on tissue NGS findings and findings from the liquid biopsies were comparable. Finally, liquid biopsy findings were, on average, reported 12 days earlier than tissue NGS findings. Collectively, these findings highlight the clinical utility of liquid biopsy and substantiate the complementary role of liquid biopsy in clinical practice.⁷

The findings from this study also highlight a role for RNA testing in liquid biopsy. Detection of gene rearrangements by ctDNA-based methods is generally challenging, with past studies demonstrating a 37%–63% sensitivity of ctDNA-based liquid biopsy relative to tissue biopsy in untreated patients with NSCLC.^{16,19,20} Although LHM ctRNA detected a numerical greater proportion of tissue NGS-confirmed G9 gene rearrangements (46.2%) than both DNA-based liquid biopsy methods (LHM ctDNA, 43.8%; G360 ctDNA, 37.5%), the sensitivity of liquid biopsy for gene rearrangements remains low. However, incorporation of ctRNA testing into LHM increased its diagnostic yield of gene rearrangements by 28.6%, demonstrating that inclusion of ctRNA improves detection of gene rearrangements. Additionally, ctRNA findings were highly specific; all ctRNA findings were orthogonally identified either by tissue NGS or by both DNA-based liquid biopsy methods. Finally, superiority of LHM ctDNA and ctRNA to G360 ctDNA was demonstrated, supporting the use of an assay that interrogates both ctDNA and ctRNA over ctDNA-only assays in clinical practice, particularly for gene rearrangements less amenable to ctDNA analysis.²¹ Indeed, although the NCCN guidelines recommend RNA testing for the detection of gene rearrangements⁵ and several studies have demonstrated the clinical value of adding tissue RNA testing to DNA,^{22–24} this study is, to our knowledge, the first to demonstrate the validity of such an approach in

liquid biopsy prospectively. However, the generalizability of this to all liquid biopsies remains to be evaluated.

This study had several interesting strengths and limitations to highlight. A key strength was the study design of a reference tissue NGS assay and two separate liquid biopsies with results independently returned to physicians in real-time, taking into consideration several challenges commonly faced in real-world practice such as tissue QNS. One limitation was that patients were followed for only 12 months after treatment initiation, whereas the median time on targeted therapy is typically longer than 12 months. Also, as the study was powered for noninferiority end points, the number of patients receiving targeted therapy in each G9 biomarker category was limited, precluding further subgroup analyses. Finally, only a subset of 101 of 151 patients underwent LHM ctRNA testing. Coupled with the low prevalence of gene rearrangements in NSCLC of 8%–17%,²⁵ this resulted in a numerically small number of gene rearrangement–positive patients identified by ctRNA. However, the improvement in diagnostic yield and demonstrated superiority of the LHM ctDNA and ctRNA assay

over G360 ctDNA indicated that this did not significantly affect the study results.

In conclusion, this multicenter, prospective, observational cohort study demonstrated that LHM ctDNA, an amplicon-based liquid biopsy, was noninferior to the FDA-approved G360 ctDNA liquid biopsy, with LHM ctDNA and G360 ctDNA detecting a tissue NGS–confirmed G9 biomarker in 72.1% and 66.2% of patients, respectively, but not to tissue NGS. Treatment outcomes based on liquid biopsy findings were comparable with those based on tissue NGS, and the addition of liquid biopsy is complementary to tissue biopsy, with liquid biopsy findings being reported faster than tissue biopsy findings and liquid biopsy identifying clinically valid and actionable biomarkers in 45.4% of cases where tissue NGS cannot be performed or in 15.2% of cases where tissue NGS returns a negative result. The incorporation of additional ctRNA testing into ctDNA liquid biopsy improves its diagnostic yield, potentially bridging the gap in sensitivity between tissue and liquid biopsies, warranting further investigations into the utility of plasma ctRNA testing in clinical practice.

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